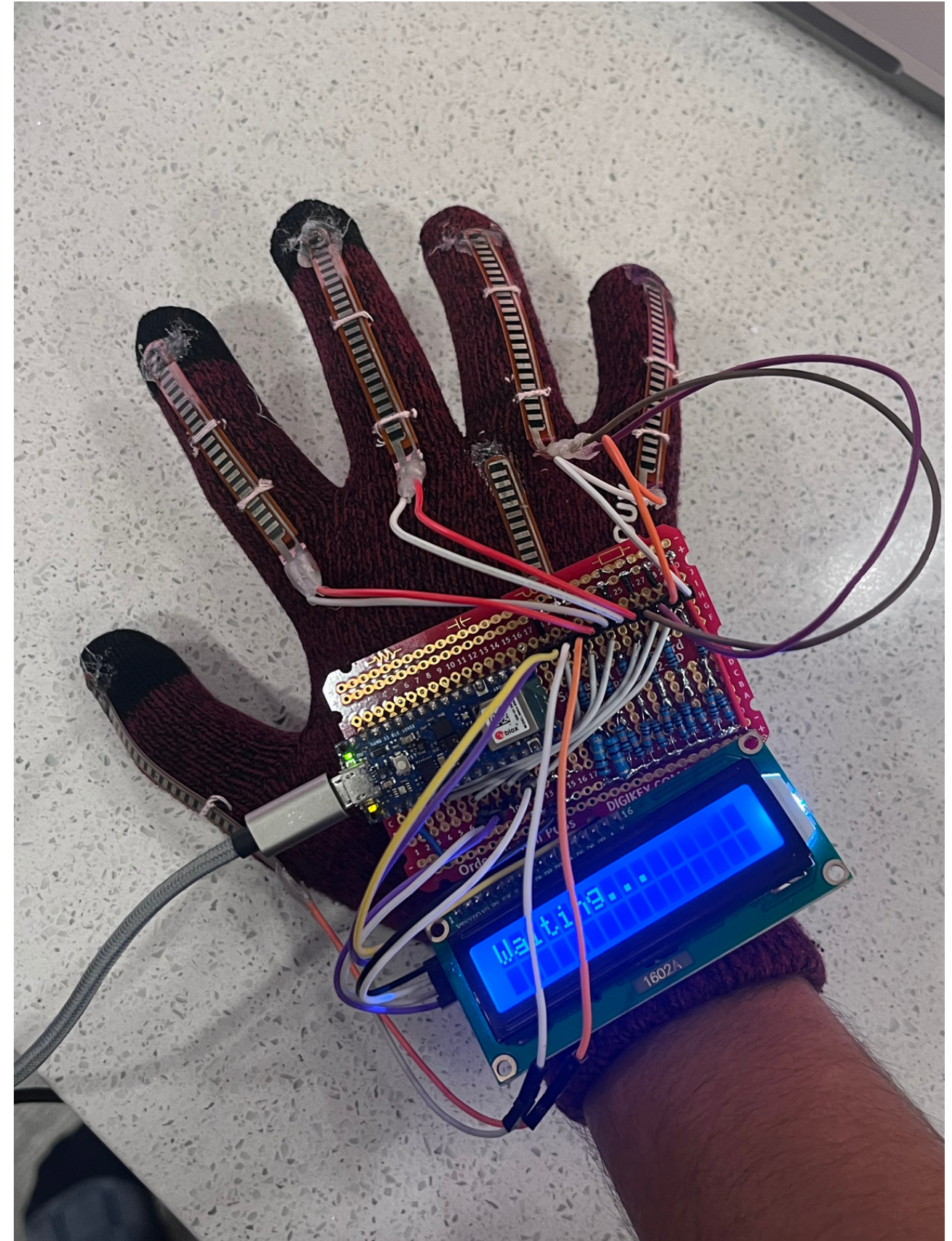
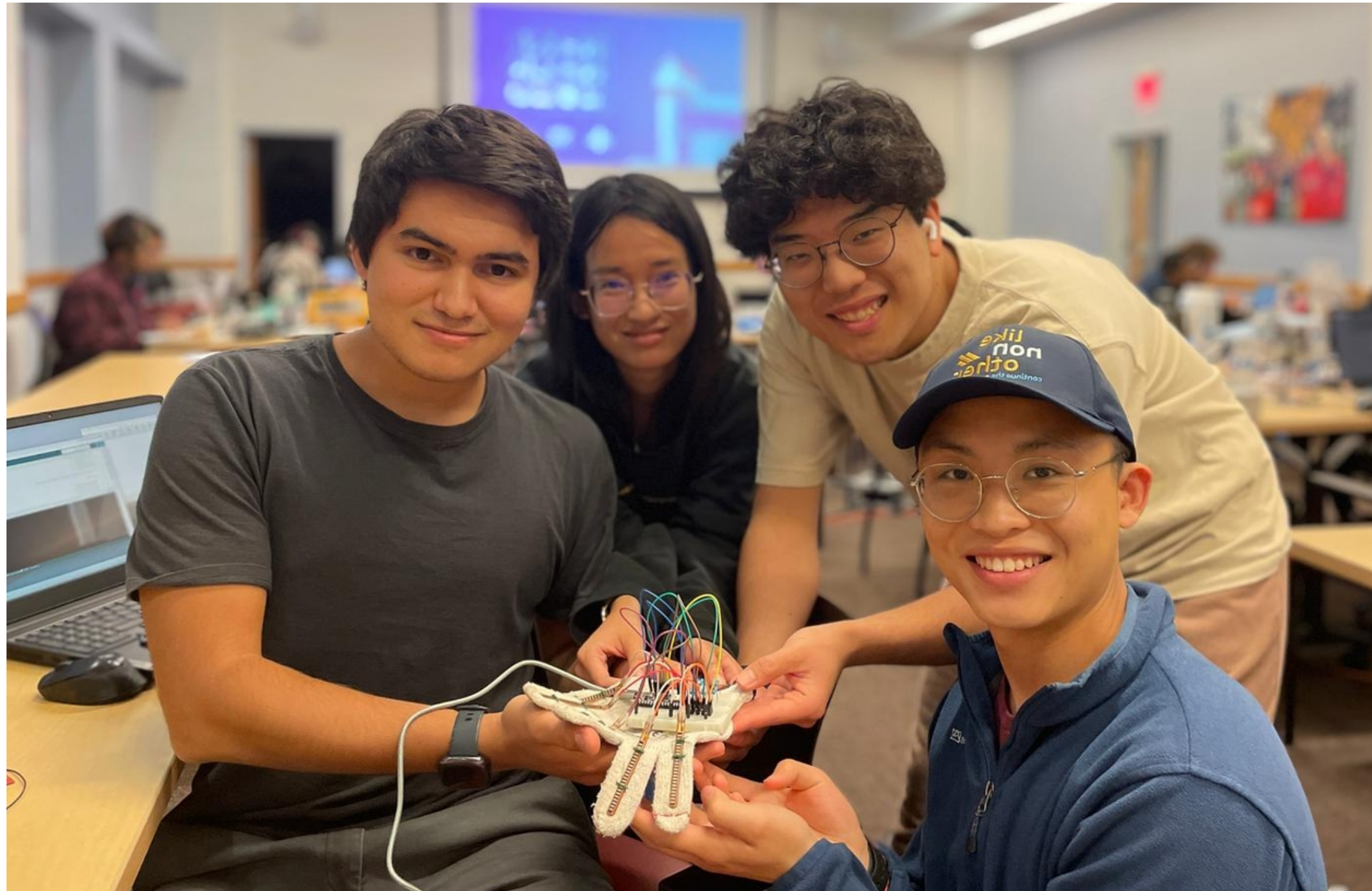


ASL GLOVE

Dehao Lin
Giovanni Bernal Ramirez
Li-Pin Chang
So Hirota



TEAM



Giovanni Bernal Ramirez | Li-Pin Chang | So Hirota | Dehao Lin

MechE

MechE

DCS

MechE

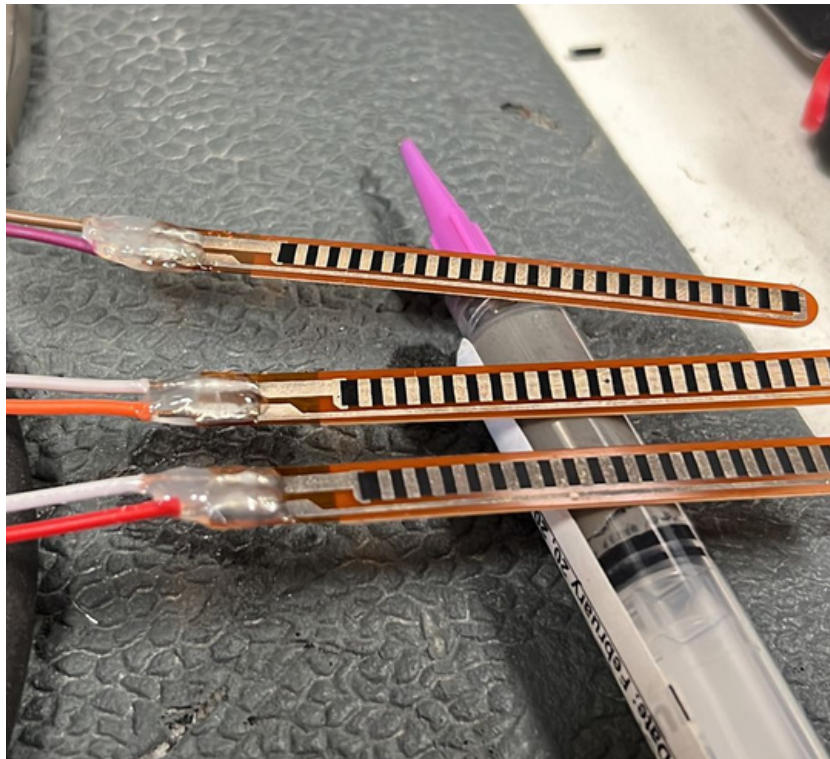
OVERVIEW

A glove that can transcribe American Sign Language alphabets and display on a cellphone App and an LCD screen through Bluetooth Low Energy.

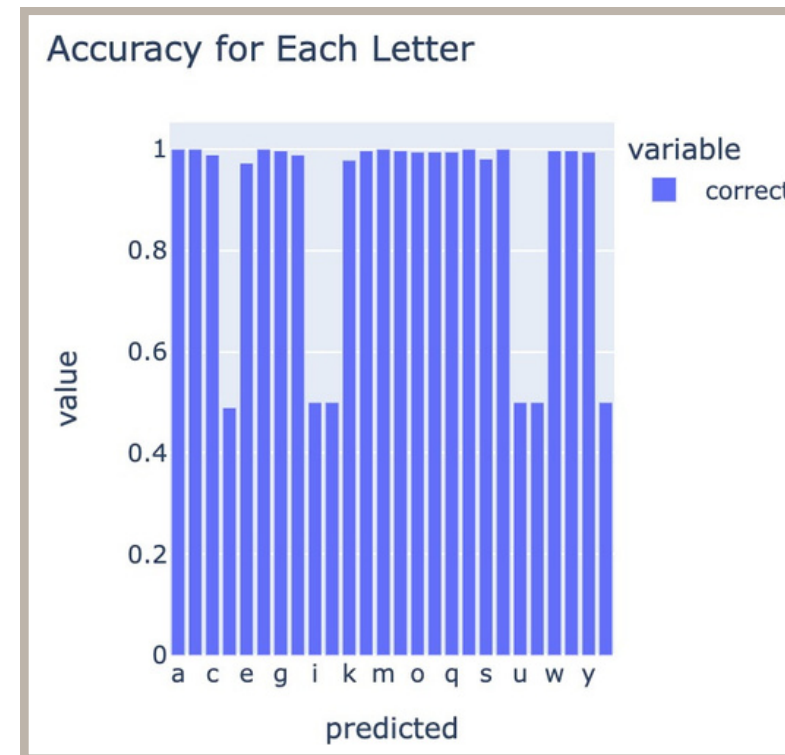
1. Improve general communication for the deaf community
2. Students to understand ASL
3. Practice ASL gesture



KEY FEATURES



Flex Sensor

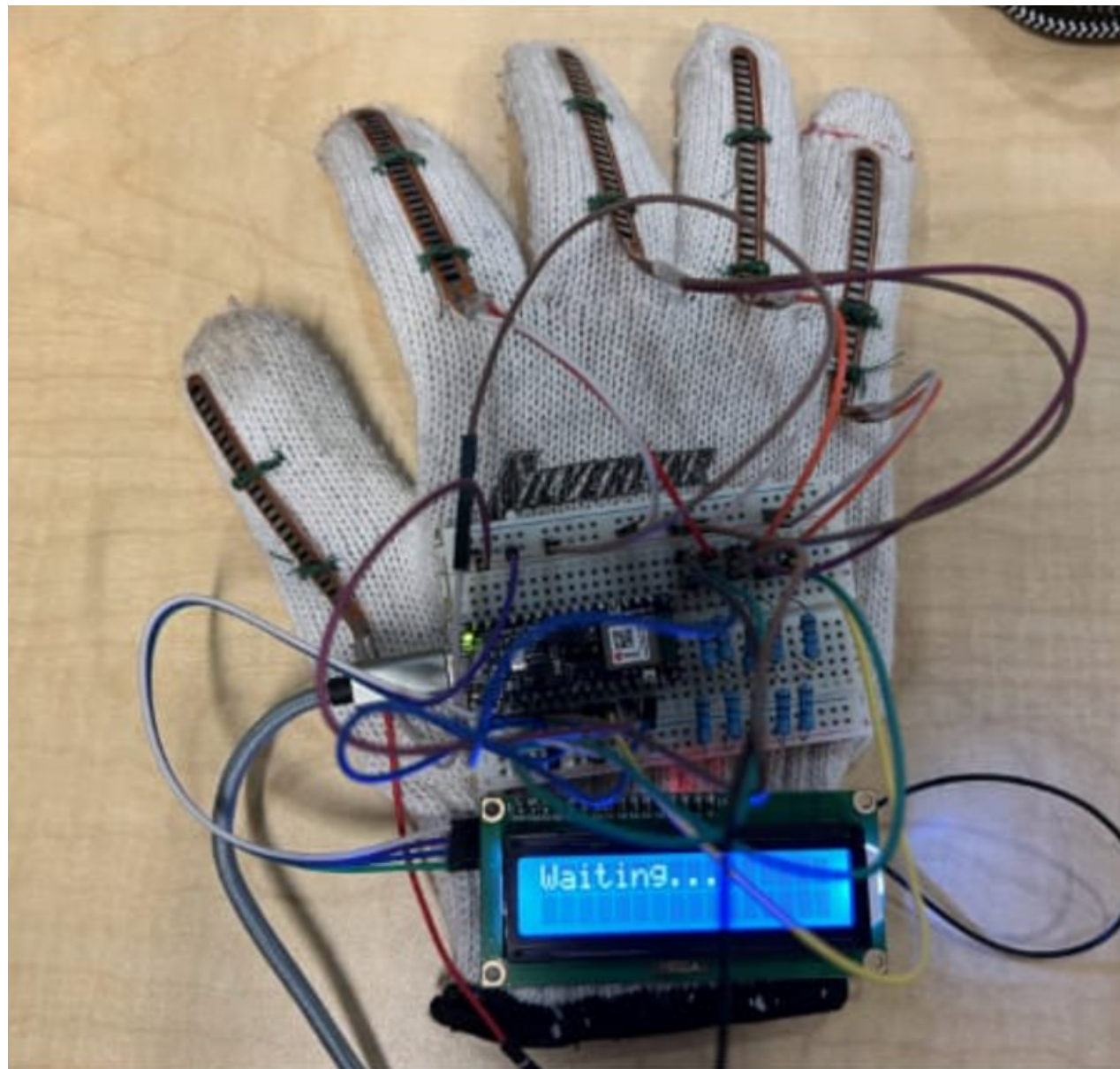


KNN Algorithm



Mobile App | BLE

ITERATIONS



Version 1, created during HardHack

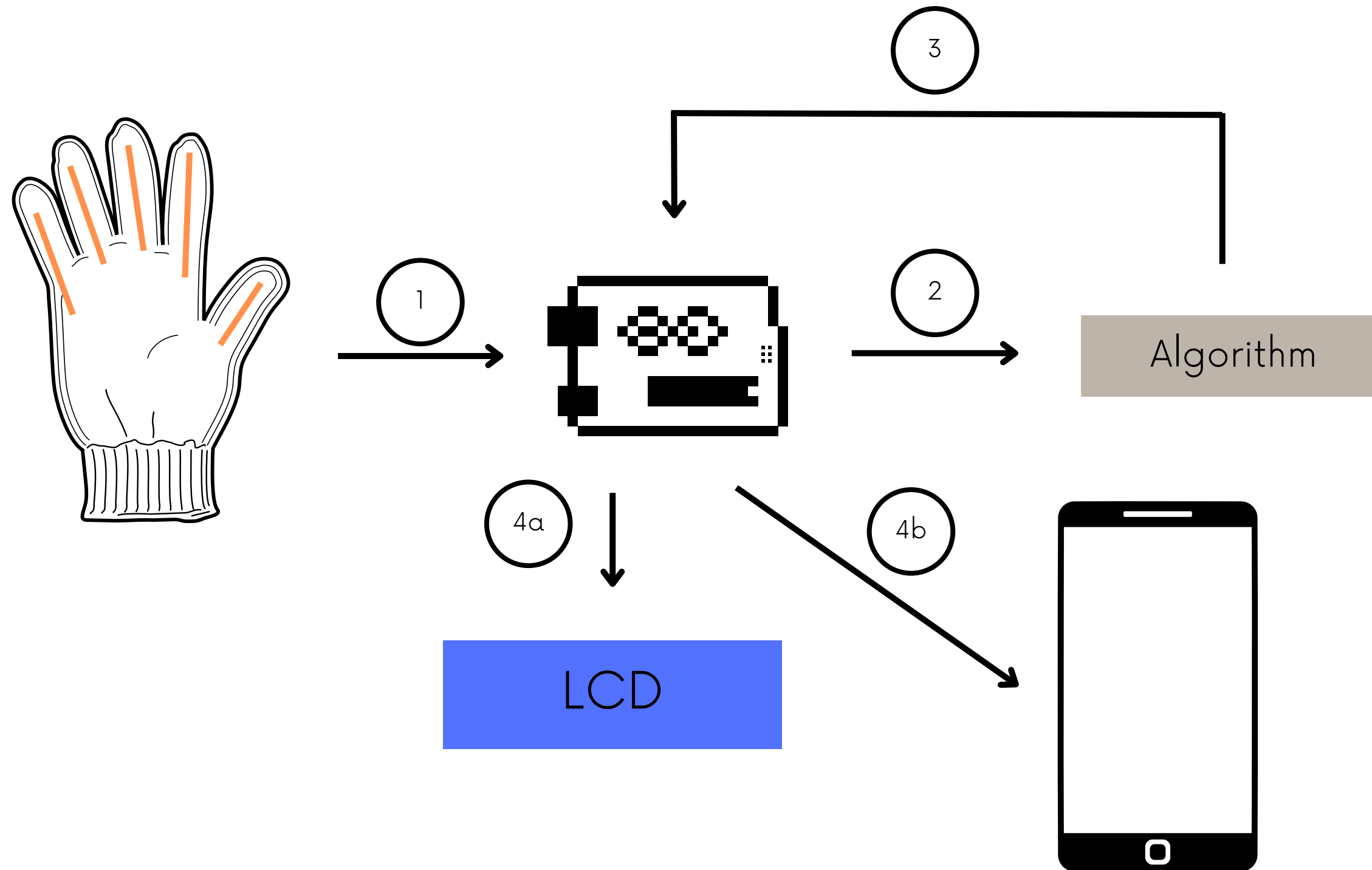


Version 2

IMPROVEMENTS

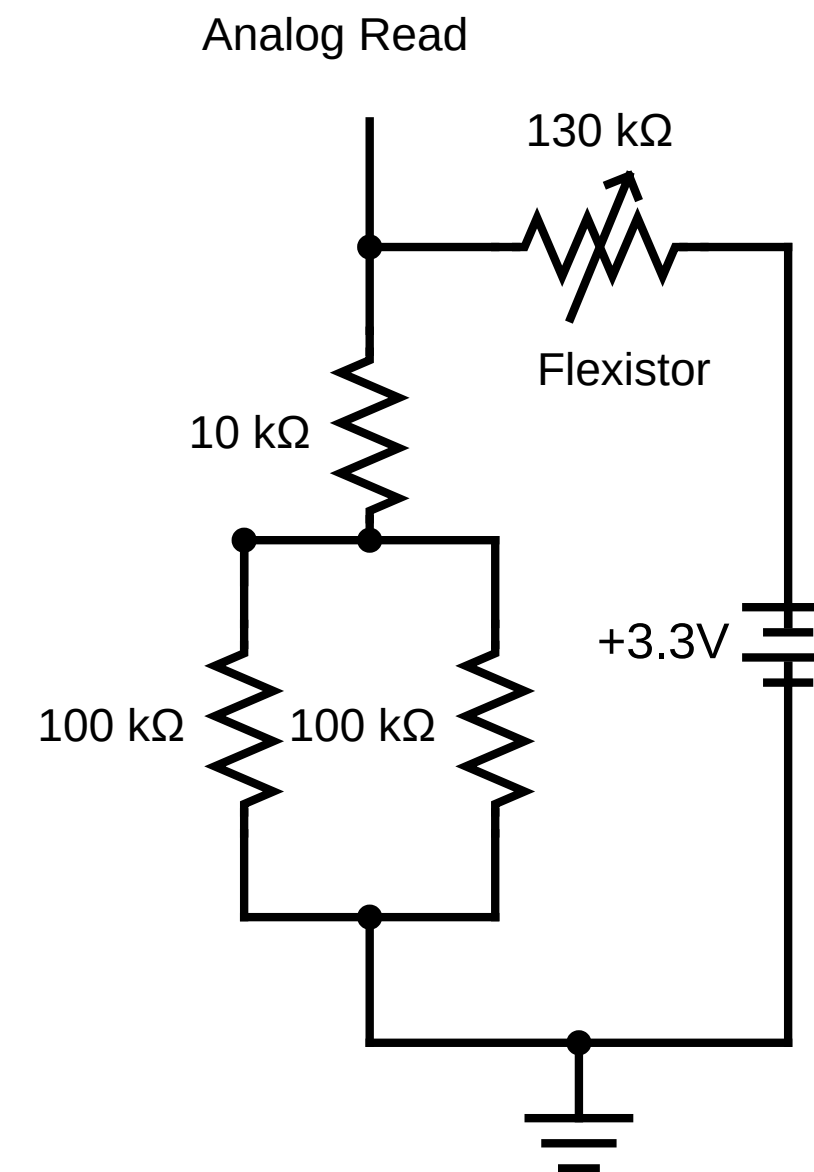
- Better fitted glove
- Soldered components onto protoboard
- Added a sixth flex sensor to get curvature of back of hand (for letters like e, m, n)
- Incorporated different commands (append, backspace)
 - Gyroscope and accelerometer integration
- Improved software
 - Increased fault tolerance

FLOW



SCHEMATICS

We use the Arduino Analog pins on a voltage divider to read the voltage across the flex sensor to measure the amount of bend in each finger.



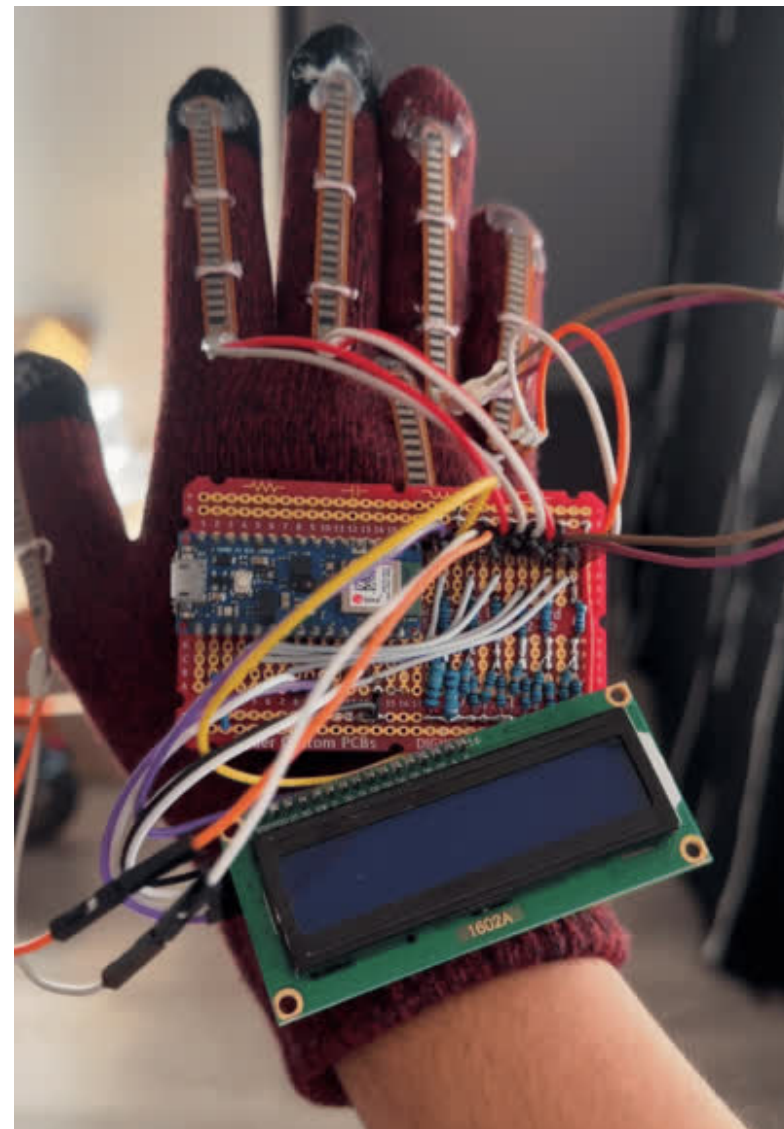
FINDING THE OPTIMAL RESISTANCE

We calculated the optimal fixed resistance between the analog pins and ground that would produce the greatest range of values. Getting a wide range of values enables the algorithm to make more precise predictions.

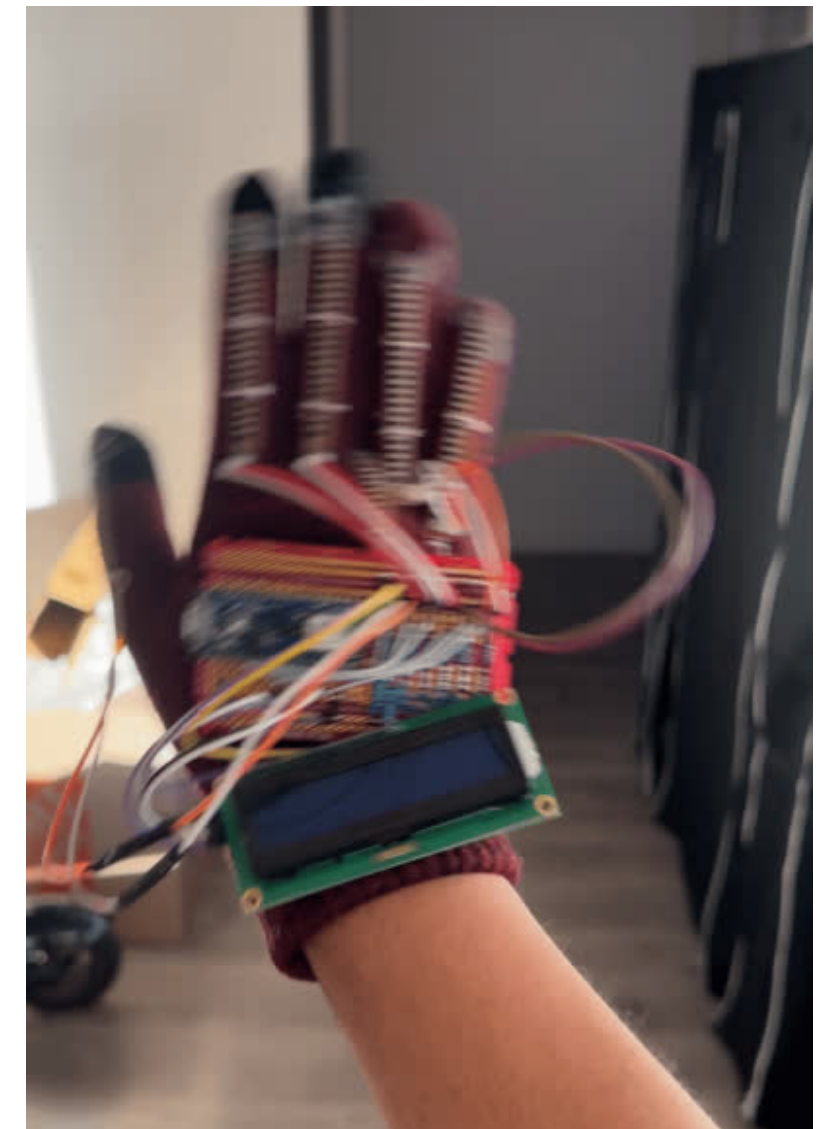


COMMANDS

Gyroscopic data was sent to the laptop, alongside the flex sensor data. We set thresholds for the gyroscopic data to trigger these commands.



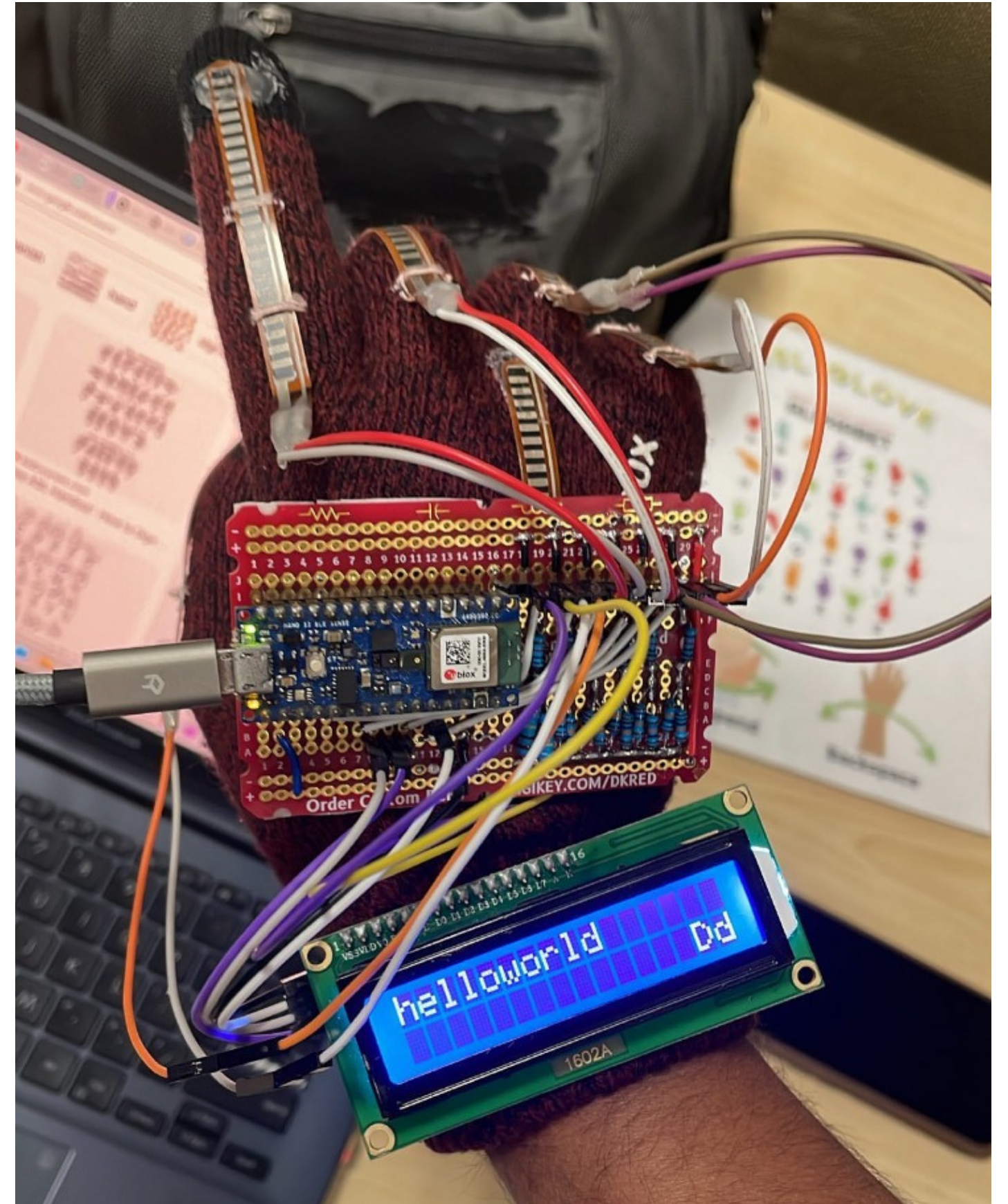
Append



Backspace

CAPABILITIES

Hello World!



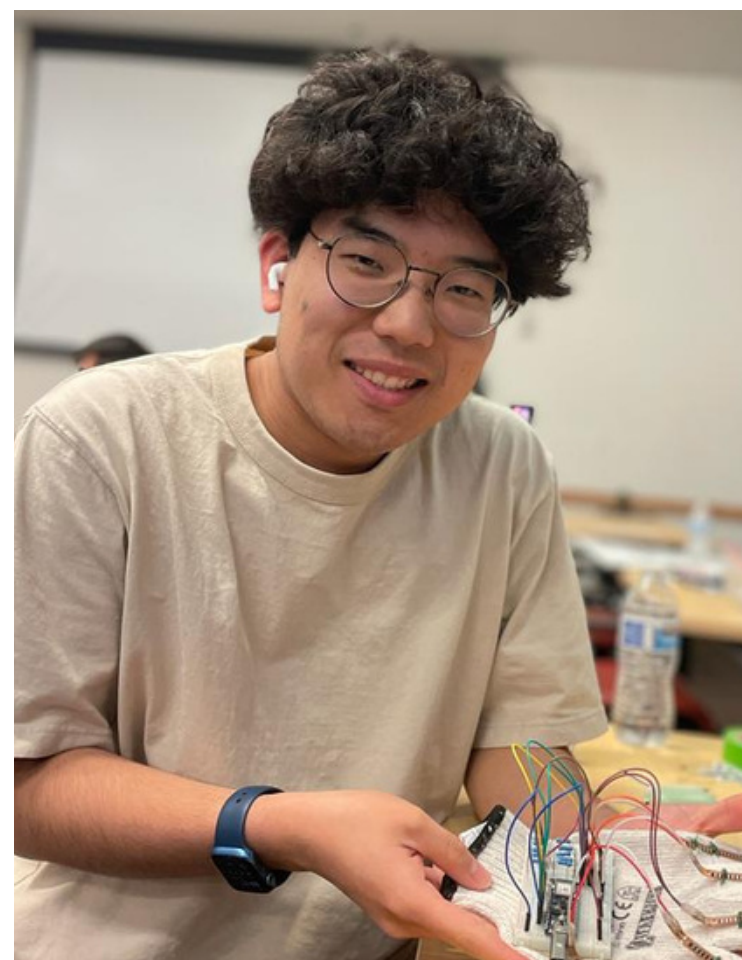
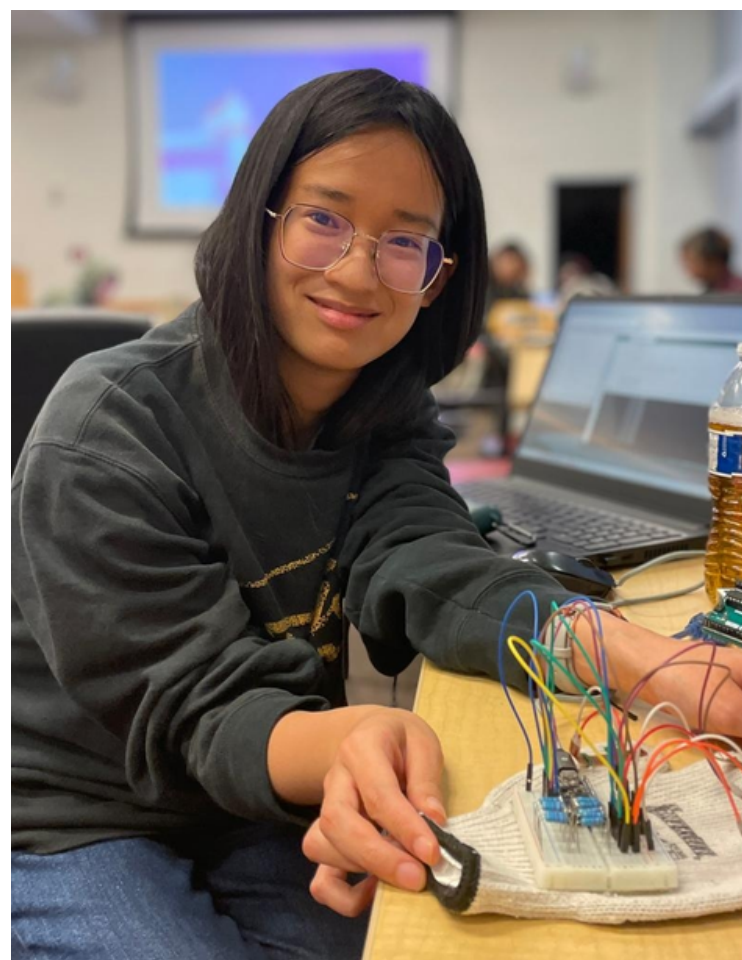
DATA COLLECTION

- Each person took turn wearing the glove and signed each alphabet for 1 minute.
- Used PuTTY to save sensor data as a csv file



FOR THE FUTURE

- Adding additional IMU data to allow for recognition of words
- Adding additional sensors to improve reliability of recognizing certain letters
 - hall effect for u, v



THANK YOU